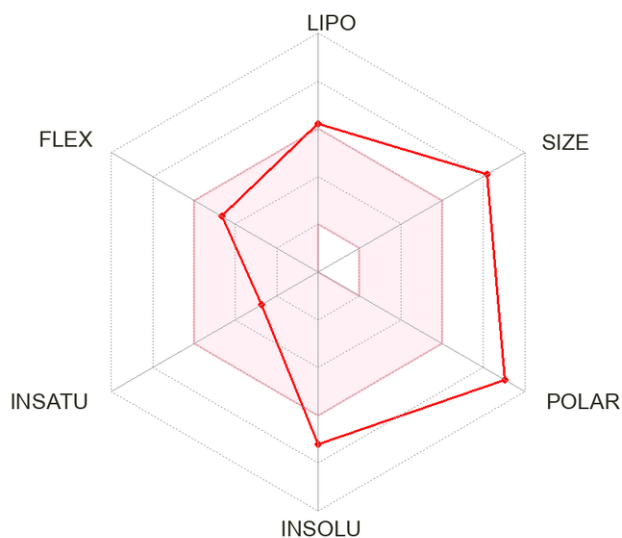
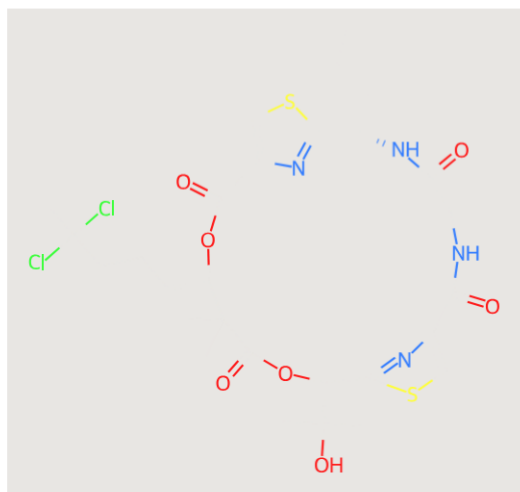


SwissADME Report

Compound: Lyngbyabellin A | **Formula:** C₂₉H₄₀Cl₂N₄O₇S₂

Source: SwissADME (swissadme.ch) — ADMET profiling, BRCA2 docking project



SMILES: CC[C@H](C)[C@H]1C2=NC(=CS2)C(=O)O[C@H](C(C(=O)O[C@H](C3=NC(=CS3)C(=O)NCC(=O)N1)C(C)(C)O)(C)C)CCCC(C)(Cl)Cl

Physicochemical Properties

Molecular weight	691.69 g/mol
Formula	C ₂₉ H ₄₀ Cl ₂ N ₄ O ₇ S ₂
Num. heavy atoms	44
Num. arom. heavy atoms	10
Fraction Csp ³	0.66
Num. rotatable bonds	7
Num. H-bond acceptors	9
Num. H-bond donors	3
Molar Refractivity	178.61
TPSA	213.29 Å ²

Lipophilicity

Log Po/w (iLOGP)	Temporarily unavailable
Log Po/w (XLOGP3)	5.38
Log Po/w (WLOGP)	4.11
Log Po/w (MLOGP)	1.19
Log Po/w (SILICOS-IT)	6.59
Consensus Log Po/w	4.32

Water Solubility

Log S (ESOL)	-7.22
Solubility (ESOL)	4.13e-05 mg/ml ; 5.97e-08 mol/l
Class (ESOL)	Poorly soluble

Log S (Ali)	-9.61
Solubility (Ali)	1.69e-07 mg/ml ; 2.44e-10 mol/l
Class (Ali)	Poorly soluble
Log S (SILICOS-IT)	-8.35
Solubility (SILICOS-IT)	3.09e-06 mg/ml ; 4.47e-09 mol/l
Class (SILICOS-IT)	Poorly soluble

Pharmacokinetics	
GI absorption	Low
BBB permeant	No
P-gp substrate	Yes
CYP1A2 inhibitor	Temporarily unavailable
CYP2C19 inhibitor	Temporarily unavailable
CYP2C9 inhibitor	Temporarily unavailable
CYP2D6 inhibitor	Temporarily unavailable
CYP3A4 inhibitor	Temporarily unavailable
Log Kp (skin permeation)	-6.70 cm/s

Druglikeness	
Lipinski	No; 2 violations: MW>500, NorO>10
Ghose	No; 3 violations: MW>480, MR>130, #atoms>70
Veber	No; 1 violation: TPSA>140
Egan	No; 1 violation: TPSA>131.6
Muegge	No; 3 violations: MW>600, XLOGP3>5, TPSA>150
Bioavailability Score	0.17

Medicinal Chemistry	
PAINS	0 alert
Brenk	2 alerts: alkyl_halide, more_than_2_esters
Leadlikeness	No; 2 violations: MW>350, XLOGP3>3.5
Synthetic accessibility	7.78